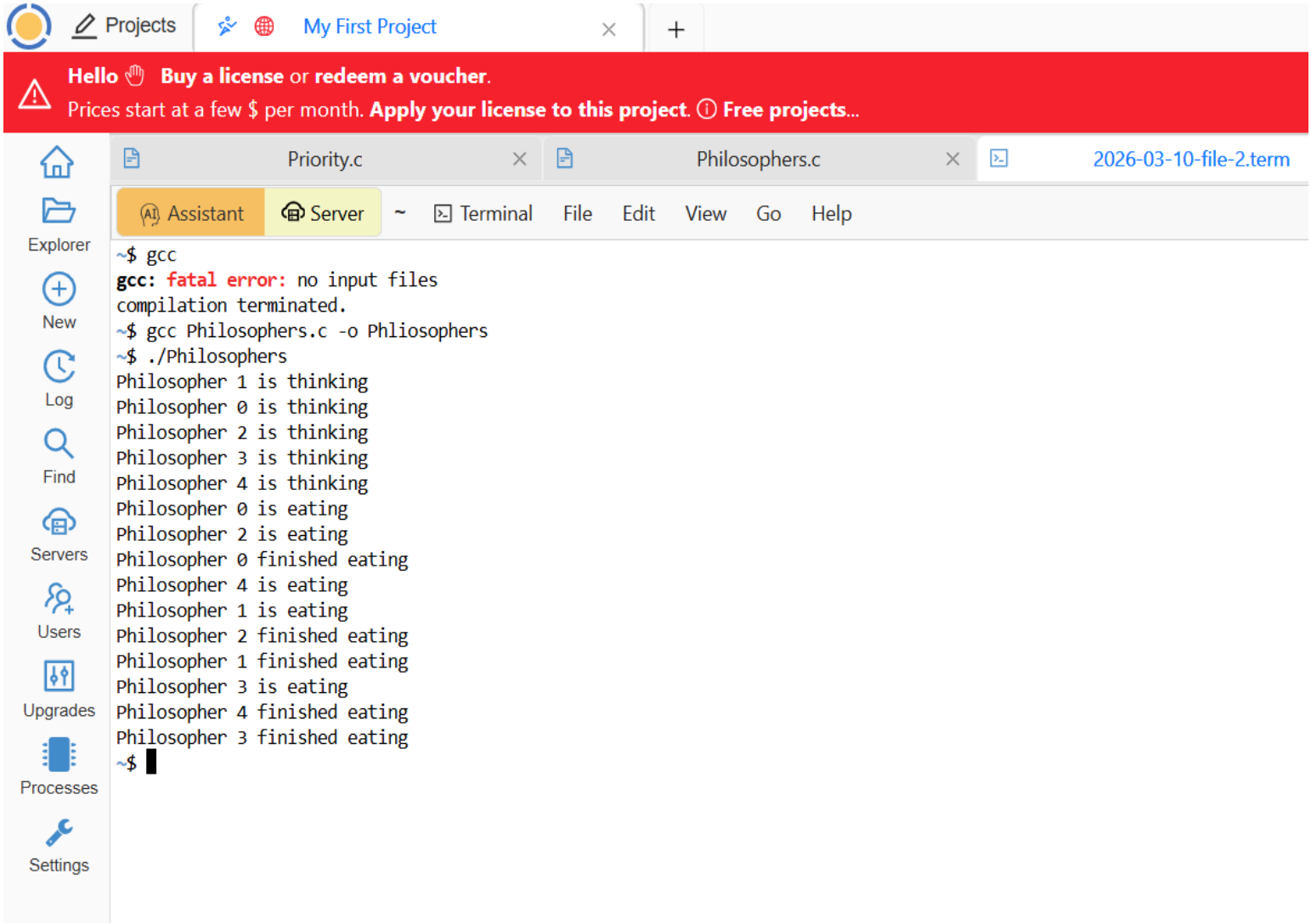
EXPERIMENT-4

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define N 5
sem_t forks[N];
pthread_t philosophers[N];
void* philosopher(void* num) {
  int id = *(int*)num;
  printf("Philosopher %d is thinking\n", id);
  sleep(1);
  sem_wait(&forks[id]);
  sem_wait(&forks[(id + 1) % N]);
    printf("Philosopher %d is eating\n", id);
  sleep(1);
  sem_post(&forks[id]);
  sem_post(&forks[(id + 1) % N]);
  printf("Philosopher %d finished eating\n", id);
  return NULL;
}
int main() {
  int i, id[N];

   for (i = 0; i < N; i++)
     sem_init(&forks[i], 0, 1);
  for (i = 0; i < N; i++) {
    id[i] = i;
    pthread_create(&philosophers[i], NULL, philosopher, &id[i]);
  }
```

```
    for (i = 0; i < N; i++)
      pthread_join(philosophers[i], NULL);
    return 0;
}
```